

Beamforming and Load-Balanced User Association in RIS-Aided mmWave Systems via Adaptive Attention Graph Neural Networks

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Abstract—This paper investigates the joint optimization of user association, base station (BS) beamforming, and reconfigurable intelligent surface (RIS) phase adjustment in an RIS-aided multi-BS multi-user-equipment (UE) millimeter wave (mmWave) network with load-balancing considerations. To address this complex problem, we propose a graph neural network (GNN)-based approach with enhanced generalizability to both varying numbers of BSs and UEs. An adaptive attention-based aggregation (AAA) mechanism is incorporated to mitigate oversmoothing in deep GNNs. By using uplink pilots as input, the proposed method avoids the need for explicit channel state information (CSI). Simulation results demonstrate the proposed scheme's superior sum-rate performance compared to both optimization-based and learning-based benchmarks while satisfying load-balancing requirements, quantify the effectiveness of AAA in addressing oversmoothing, demonstrate the proposed scheme's robustness to pilot correlation, and highlight the load-balancing benefits of deploying RISs.

Index Terms—Reconfigurable intelligent surface, mmWave, beamforming, user association, load balancing, deep learning, graph neural network.

I. INTRODUCTION

RECONFIGURABLE intelligent surfaces (RISs) are considered to be a promising technology for expanding coverage and enhancing spectral efficiency in wireless systems [2], [3], [4]. RISs enable the reconfiguration of the transmission environment to mitigate signal blockage and interference. Thus, RISs have the potential to improve the spectral efficiency of mmWave communications that suffer

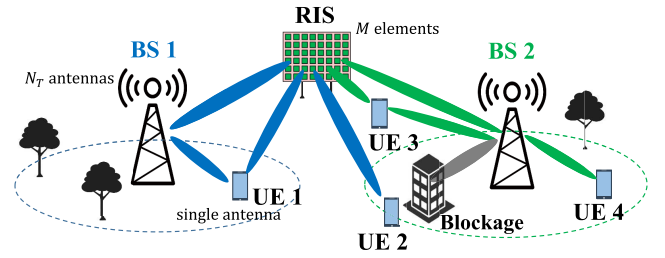


Fig. 1. An RIS-assisted multi-BS multi-UE system (an example of two BSs and 4 UEs is shown here).

serious signal attenuation. Various methods have been proposed to jointly optimize transmit beamforming and RIS phase shifts, employing conventional optimization techniques (e.g., [2], [3]), data-driven deep learning approaches (e.g., [5], [6], [7]), or graph neural network (GNN) techniques (e.g., [8], [9], [10], [11]). Among these methods, GNNs have proven their viability in achieving superior performance and lower complexity compared to optimization-based approaches, as well as enhanced generalizability compared to deep neural network (DNN)-based approaches, as shown in [8] and [9].

Another advantage of RIS-aided systems is the enhanced flexibility in *user association (UA)* due to coverage extension. For example, as illustrated in Fig. 1, user equipment (UE) 2, which is not within the coverage of base station (BS) 1, may now be associated with BS 1 with the assistance of an RIS. This proves valuable when the link between UE 2 and BS 2 is obstructed, even if UE 2 is closer to BS 2. Another interesting observation is that the UA strategy becomes less straightforward when *load balancing* is taken into consideration. For the example in Fig. 1, when considering load distribution among BSs, a more balanced UA strategy may link UE 2 to the less-loaded BS 1 instead of BS 2, even though linking UE 2 to BS 2 is also feasible with the assistance of an RIS. Therefore, RIS technology has the potential to facilitate a more flexible and load-aware UA strategy. RIS-assisted UA in multi-BS multi-UE systems have been studied in [12], [13], and [14] using conventional optimization techniques and alternating optimization (AO) methods. Mei and Zhang [12] proposed a joint design of RIS-UE association and BS power control to balance signal-to-interference-plus-noise ratio (SINR) among

Received 11 February 2025; revised 15 August 2025; accepted 9 October 2025. Date of publication 21 October 2025; date of current version 22 December 2025. This work was supported in part by the National Science and Technology Council, Taiwan, under Grant NSTC 112-2221-E-A49-152-MY3 and Grant NSTC 112-2221-E-001-004-MY3; in part by the MediaTek Advanced Research Center; and in part by an Innovation Grant from Princeton NextG. An earlier version of this paper was presented in part at the IEEE International Conference on Communications (ICC), Denver, CO, USA, June 9–13, 2024 [DOI: 10.1109/ICC51166.2024.10622705]. The associate editor coordinating the review of this article and approving it for publication was M. Kaneko. (Corresponding authors: Ronald Y. Chang; Feng-Tsun Chien.)

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Digital Object Identifier 10.1109/TWC.2025.3621268

UEs in multi-RIS-aided systems. Zhao et al. [13] aimed at jointly optimizing BS-UE association, BS power control, and RIS phase shifts in RIS-aided systems. Taghavi et al. [14] addressed the joint optimization of BS active beamforming, RIS passive beamforming, and BS-UE association to maximize the sum rate in multi-RIS-aided systems, and proposed an AO-based algorithm which requires multiple iterations. However, conventional optimization and AO-based algorithms have high complexity and long execution times.

To overcome these limitations, learning-based methods such as DNNs [15] and deep reinforcement learning (DRL) [16], [17], [18] have been proposed for RIS-assisted UA. Amirara et al. [15] proposed a DNN-based approach to jointly design RIS-user association, RIS reflection matrix, and BS beamforming for a system aided by multiple RISs. Aung et al. [16] proposed a DRL-based approach to maximize energy efficiency in multi-RIS networks by jointly optimizing RIS-user association, reflective elements ON/OFF states, phase shift, and transmit power. Wang et al. [17] proposed a Deep Q-Network (DQN)-based algorithm to jointly optimize BS-user association, computation offloading, resource allocation, and RIS phase shifts for minimizing long-term energy consumption. Zhu et al. [18] proposed a soft actor-critic DRL-based approach to jointly optimize the BS-RIS-UE association and RIS passive beamforming, aiming to maximize the sum rate in a RIS-assisted wireless network. Nevertheless, DNN and DRL approaches for RIS-assisted UA, as proposed in [15], [16], [17], and [18], lack generalizability across different system configurations, such as varying numbers of UEs. Specifically, the number of parameters in the output layer of a DNN or the action network of a DRL depends on the number of UEs and/or BSs. As a result, any change in the number of UEs or BSs necessitates retraining the DNN or DRL model, limiting their adaptability to dynamic network settings.

GNNs are a promising approach to address generalizability challenges [8], [9], [19]. GNNs operate on a graph structure through two key processes: aggregation, where nodes exchange information, and combination, where gathered information is integrated and updated via neural networks. Through aggregation and combination of information of nodes, GNN has the potential to achieve effective learning outcomes for communication problems. Moreover, GNN's *permutation invariance and equivariance properties* enable effective generalization to diverse system settings (such as numbers of UEs and BSs) without the need for retraining the embedded model. Permutation invariance means that the overall output remains unchanged regardless of node ordering, while permutation equivariance means that permuting the input results in the corresponding permutation of the output. GNN-based approaches have been proposed for designing UA strategies in multi-BS systems [20], [21], [22]. Randall et al. [20] proposed combining Double Deep Q Networks with GNNs to develop a UA algorithm aimed at maximizing network throughput while ensuring user fairness in small-cell networks. Zhang et al. [21] proposed a digital twin architecture integrated with a GNN to jointly optimize UA and power allocation to maximize the average user rate in terahertz multi-cell networks.

Ranasinghe et al. [22] presented a GNN-based access point (AP) selection algorithm based on path loss and shadow fading for cell-free massive multiple-input multiple-output (MIMO) systems. These studies [20], [21], [22] however did not consider RIS-aided scenarios or incorporate load balancing into their UA strategy designs. While user association in RIS-aided systems was investigated in [23], the GNN-based RIS-UE association strategy proposed for single-BS multi-UE systems cannot be directly applied to multi-BS scenarios.

GNN-based approaches have been proposed for the joint design of transmit beamforming and RIS phase shifts [8], [9], [10], [11]. Jiang et al. [8] developed a GNN architecture parameterizing the mapping from received pilots to BS and RIS beamforming without explicit channel estimation. Chen et al. [9] proposed a two-phase GNN for sum rate maximization in multi-UE MISO systems with a relay and RIS. Jin et al. [10] introduced a model-driven GNN to reduce the complexity of joint BS and RIS beamforming. Yeh et al. [11] incorporated angular CSI and graph attention to jointly design BS beamforming and RIS phase shifts for multi-UE MISO mmWave networks. However, these works focused on single-BS systems and cannot be directly extended to multi-BS scenarios or to problems involving UA which is specific to multi-BS settings.

This paper addresses this research gap by jointly optimizing UA, transmit beamforming (BF), and RIS phase shifts in a RIS-assisted multi-BS multi-UE mmWave system with BS load balancing considerations, through a GNN-based approach extended from our prior work [1]. This joint design is challenging due to the non-convex nature of the problem and the strong coupling among the variables, especially in multi-BS settings. Furthermore, in mmWave systems, accurate CSI acquisition is difficult. GNNs offer a promising solution by learning design variables directly from pilot signals, bypassing explicit channel estimation and significantly reducing computational burden compared to traditional AO-based methods. The proposed GNN-based method introduces two unique features. First, while most existing GNN-based methods [20], [21], [22], [24], [25] demonstrate GNN's generalizability to the number of UEs, our model is designed with inherent generalization capabilities for *both* the number of UEs and BSs. Second, the proposed GNN includes a built-in mechanism to handle the *oversmoothing* issue [26] inherent in deep GNNs. Oversmoothing refers to the issue where graph node features become increasingly similar after passing through a deep updating structure, which could potentially degrade a GNN's learning capability. We introduce an adaptive attention-based aggregation (AAA) mechanism that integrates adaptivity into GNNs, enabling adaptive control of node information exchange to strike a balance between mitigating oversmoothing and maintaining learning effectiveness. This approach is novel to the considered problem and distinguishes it from previous applications such as urban traffic flow forecasting [27] and fraud detection [28]. The main contributions of this paper are summarized as follows:

- We propose a GNN-based approach to jointly optimize UA, BS transmit beamforming, and RIS phase shifts in RIS-aided multi-BS multi-UE systems, while

TABLE I
SUMMARY OF RELATED WORKS

Reference	Problem	Load Balancing	Generalizability	Handling Oversmoothing	Method
[12]	RIS-UE association, BS power control	×	×	×	AO
[13]	BS-UE association, BS power control, RIS phase shifts	×	×	×	AO
[14]	BS-UE association, transmit BF, RIS phase shifts	✓	×	×	AO
[15]	RIS-UE association, transmit BF, RIS phase shifts	×	×	×	DNN
[16]	RIS-UE association, On/Off states of RIS elements, RIS phase shifts, BS power control	×	×	×	DRL
[17]	BS-UE association, computation offloading, resource allocation, RIS phase shifts	×	×	×	DRL
[18]	BS-RIS-UE association, RIS phase shifts	×	×	×	DRL
[1]	BS-UE association, transmit BF, RIS phase shifts	✓	number of UEs	×	GNN
This work	BS-UE association, transmit BF, RIS phase shifts	✓	number of BSs & UEs	✓	GNN

TABLE II
LIST OF SYMBOLS

Number of BSs	B
Number of transmit antennas at each BS	N_T
Number of UEs	U
Number of RIS elements	M
The RIS phase shift matrix	Φ
The m th diagonal element of Φ	ϕ_m
The BS-UE association indicator	$\delta_{b,u}$
The set of BSs serving the u th UE	$\mathcal{S}_u^{\text{BS}}$
The set of UEs served by the b th BS	$\mathcal{C}_b^{\text{UE}}$
The channel between the b th BS and the RIS	$\mathbf{H}_{b,R}$
The channel between the RIS and the u th UE	$\mathbf{h}_{R,u}$
The channel between the b th BS and the u th UE	$\mathbf{h}_{b,u}$
The beamforming vector of the (b,u) th BS-UE link	$\mathbf{f}_{b,u}$
The achievable rate of the u th UE	R_u
The maximum number of UEs served by the b th BS	$Z_{\max,b}$
The maximum transmit power at the b th BS	$P_{\max,b}$
Length of pilot signals	L
Number of updating layers	Q
The feature extractor for BS-UE nodes in initial layer	$\Psi_U^{(0)}$
The feature extractor for the RIS node in initial layer	$\Psi_R^{(0)}$
The aggregator for BS-UE nodes in τ th updating layer	$\Psi_U^{\text{ag},(\tau)}$
The aggregator for the RIS node in τ th updating layer	$\Psi_R^{\text{ag},(\tau)}$
The combiner for BS-UE nodes in τ th updating layer	$\Psi_U^{\text{cb},(\tau)}$
The combiner for the RIS node in τ th updating layer	$\Psi_R^{\text{cb},(\tau)}$
Parameter of feature extractors, aggregators, and combiners	$\mathcal{K}$
The fully-connected linear layer for UA	$\Psi_{\text{UA}}(\cdot)$
The fully-connected linear layer for BF	$\Psi_{\text{BF}}(\cdot)$
The fully-connected linear layer for RIS phase shifts	$\Psi_{\text{RIS}}(\cdot)$
The initial feature of the (b,u) th BS-UE node	$\mathbf{z}_{(b,u)}^{(1)}$
The initial feature of the RIS node	$\mathbf{z}_R^{(1)}$
The (b,u) th BS-UE node feature in τ th updating layer	$\mathbf{z}_{(b,u)}^{(\tau)}$
The (b,u) th BS-UE aggregated feature in τ th updating layer	$\mathbf{z}_{(b,u)}^{\text{ag},(\tau)}$
The RIS aggregated feature in τ th updating layer	$\mathbf{z}_R^{\text{ag},(\tau)}$
The RIS node feature in τ th updating layer	$\mathbf{z}_R^{(\tau)}$
The output feature for UA	$\mathbf{z}_{(b,u)}^{\text{out}}$
The output feature for BF	$\mathbf{z}_{(b,u)}^{\text{BF}}$
The output feature for RIS phase shifts	$\mathbf{z}^{\text{RIS}}$

satisfying load-balancing constraints. The proposed method avoids the need for explicit channel estimation by learning directly from pilot signals. It demonstrates generalizability to varying numbers of UEs and BSs, enabling low-complexity training with few UEs and BSs and allowing the direct use of shared, pre-trained GNN modules for newly added BS-UE links without requiring retraining. The proposed GNN model employs unsupervised training to maintain low complexity.

- We propose a built-in mechanism called AAA within the proposed GNN model, consisting of two components: 1) sparse attention for aggregation and 2) tuning aggregation for combination. This mechanism mitigates oversmoothing in deep GNNs by refining the node feature aggregation process, with adaptive control to balance information exchange and optimize learning performance, while adding only modest complexity relative to the overall model.
- Simulation results show that the proposed scheme achieves superior sum-rate performance compared to both optimization-based and learning-based benchmarks, with favorable complexity, high generalizability, robustness to pilot correlation, and proven effectiveness in mitigating oversmoothing. A detailed complexity analysis of the proposed scheme is also presented.

A summary of closely related works addressing similar problems is provided in Table I, which also highlights the distinct contributions and unique aspects of this paper.

The rest of this paper is organized as follows. Section II presents the system model and problem formulation. Section III describes the proposed GNN-based method. Section IV analyzes the computational complexity of the proposed method. Section V presents simulation results and discussion. Finally, Section VI provides concluding remarks and potential future directions.

Notation: $\mathbf{a}^*$, $\mathbf{a}^T$, and $\mathbf{a}^H$ represent the conjugate, transpose, and conjugate transpose of vector $\mathbf{a}$, respectively. $[\mathbf{a}]_i$ represents the i th entry of vector $\mathbf{a}$, and $[\mathbf{a}]_{i:j}$ represents the i th entry to the j th ($j > i$) entry of vector $\mathbf{a}$. $\Re\{\cdot\}$ and $\Im\{\cdot\}$ denote the real and imaginary parts of the argument, respectively. $|\cdot|$, $\|\cdot\|_0$, $\|\cdot\|$, and $\|\cdot\|_F$ denote the absolute value, 0-norm, 2-norm, and Frobenius norm, respectively. $\text{diag}\{a_1, a_2, \dots, a_N\}$ represents a diagonal matrix with terms $a_1, a_2, \dots, a_N$ on the diagonal. $\lfloor \cdot \rfloor$ denotes the floor operation.

II. SYSTEM MODEL AND PROBLEM DESCRIPTION

In this section, we present the system model and problem description. The symbols used in this paper are summarized in Table II.

A. System Model

We consider a RIS-aided downlink mmWave communication system with B BSs, each equipped with N_T transmit antennas, and U single-antenna UEs, as illustrated in Fig. 1.

The sets of all BSs and UEs are denoted by $\mathcal{B} = \{1, 2, \dots, B\}$ and $\mathcal{U} = \{1, 2, \dots, U\}$, respectively. The RIS is composed of M passive reflecting elements with adjustable continuous phase terms $\{\phi_1, \dots, \phi_M\} \triangleq \{e^{j\theta_1}, \dots, e^{j\theta_M}\}$, which rotates the incident signals and can be formulated as the diagonal phase matrix

$$\Phi = \text{diag}\{\phi_1, \phi_2, \dots, \phi_M\}. \quad (1)$$

The indicator of the BS-UE association is defined as $\delta_{b,u} \in \{0, 1\}$, for $u \in \mathcal{U}$ and $b \in \mathcal{B}$, where $\delta_{b,u} = 1$ represents that the u th UE is served by the b th BS, and $\delta_{b,u} = 0$ otherwise. $\mathcal{S}_u^{\text{BS}} = \{b \mid \delta_{b,u} = 1, b \in \mathcal{B}\}$ is the set of BSs serving the u th UE, whereas the set of UEs served by the b th BS is denoted by $\mathcal{C}_b^{\text{UE}} = \{i \mid \delta_{b,i} = 1, i \in \mathcal{U}\}$.

To model the rich scattering properties of the mmWave channel, we adopt the clustered channel model [29]. The channel between the b th BS and the RIS, denoted by $\mathbf{H}_{b,R} \in \mathbb{C}^{M \times N_T}$, the channel between the RIS and the u th UE, denoted by $\mathbf{h}_{R,u} \in \mathbb{C}^{M \times 1}$, and the channel between the b th BS and the u th UE, denoted by $\mathbf{h}_{b,u} \in \mathbb{C}^{N_T \times 1}$, are defined as follows:

$$\begin{aligned} \mathbf{H}_{b,R} &= \sqrt{\frac{N_T M}{N_\gamma}} \rho_{b,R} \sum_{\ell=1}^{N_{cl}} \sum_{i=1}^{L_\ell} \alpha_{\ell,i} \mathbf{a}_{\text{RIS}}(\psi_{\ell,i}^{R,r}, \theta_{\ell,i}^{R,r}) \\ &\quad \times \mathbf{a}_{\text{BS}}^H(\psi_{\ell,i}^{b,t}, \theta_{\ell,i}^{b,t}), \end{aligned} \quad (2)$$

$$\mathbf{h}_{R,u} = \sqrt{\frac{M}{N_\gamma}} \rho_{R,u} \sum_{\ell=1}^{N_{cl}} \sum_{i=1}^{L_\ell} \alpha_{\ell,i} \mathbf{a}_{\text{RIS}}(\psi_{\ell,i}^{R,t}, \theta_{\ell,i}^{R,t}), \quad (3)$$

$$\mathbf{h}_{b,u} = \sqrt{\frac{N_T}{N_\gamma}} \rho_{b,u} \sum_{\ell=1}^{N_{cl}} \sum_{i=1}^{L_\ell} \alpha_{\ell,i} \mathbf{a}_{\text{BS}}(\psi_{\ell,i}^{b,t}, \theta_{\ell,i}^{b,t}), \quad (4)$$

where $N_\gamma = \sum_{\ell=1}^{N_{cl}} L_\ell$, and $\alpha_{\ell,i}$ is the complex gain of the i th ray in the ℓ th cluster, which are assumed to be Gaussian distributed with zero mean and unit variance. $\rho_{b,R}$, $\rho_{R,u}$, and $\rho_{b,u}$ are the path loss of the channels $\mathbf{H}_{b,R}$, $\mathbf{h}_{R,u}$, and $\mathbf{h}_{b,u}$, respectively. The azimuth and elevation angles of departure (AoD) are denoted by $\psi_{\ell,i}^{b,t}, \theta_{\ell,i}^{b,t}$ for the b th BS and $\psi_{\ell,i}^{R,t}, \theta_{\ell,i}^{R,t}$ for the RIS. The azimuth and elevation angles of arrival (AoA) are denoted by $\psi_{\ell,i}^{R,r}, \theta_{\ell,i}^{R,r}$ for the RIS. The array response vector for each BS, configured with a uniform linear array aligned parallel to the x-axis, is expressed as $[\mathbf{a}_{\text{BS}}]_{n_t}(\psi, \theta) = e^{j \frac{2\pi n_t d^{\text{BS}}}{\lambda_c} \cos(\psi) \cos(\theta)}$ for $n_t = 1, \dots, N_T$, where we set the antenna spacing $d^{\text{BS}} = \lambda_c/2$. The array response vector for the RIS, configured with a uniform rectangular array placed on the yz-plane, is expressed as $[\mathbf{a}_{\text{RIS}}(\psi, \theta)]_n = e^{j \frac{2\pi d^{\text{RIS}}}{\lambda_c} (n_1 \sin(\psi) \cos(\theta) + n_2 \sin(\theta))}$, where $n_1 = \text{mod}(n-1, \sqrt{M})$, $n_2 = \lfloor (n-1)/\sqrt{M} \rfloor$, and we set the RIS element spacing $d^{\text{RIS}} = \lambda_c/2$.

The received signal of the u th UE is given by

$$\begin{aligned} y_u &= \sum_{b \in \mathcal{S}_u^{\text{BS}}} (\mathbf{h}_{b,u}^H + \mathbf{h}_{R,u}^H \Phi \mathbf{H}_{b,R}) \mathbf{f}_{b,u} s_u \\ &\quad + \sum_{b \in \mathcal{S}_u^{\text{BS}}} \sum_{v \in \mathcal{C}_b^{\text{UE}}, v \neq u} (\mathbf{h}_{b,u}^H + \mathbf{h}_{R,u}^H \Phi \mathbf{H}_{b,R}) \mathbf{f}_{b,v} s_v \\ &\quad + \sum_{\ell \in \mathcal{B} \setminus \mathcal{S}_u^{\text{BS}}} \sum_{v \in \mathcal{C}_\ell^{\text{UE}}} (\mathbf{h}_{\ell,u}^H + \mathbf{h}_{R,u}^H \Phi \mathbf{H}_{\ell,R}) \mathbf{f}_{\ell,v} s_v + n_u, \end{aligned} \quad (5)$$

where s_u is the u th UE's data symbol with $\mathbb{E}(s_u) = 0$, $\mathbb{E}(|s_u|^2) = 1$, and $\mathbb{E}(s_u^H s_v) = 0, \forall v \neq u$, $\mathbf{f}_{b,u} \in \mathbb{C}^{N_T \times 1}$ is the beamformer at the b th BS for the u th UE, and n_u is the additive white Gaussian noise (AWGN) with zero mean and variance σ^2 . The second and third terms on the right-hand side of (5) correspond to the intra-cell and inter-cell interference, respectively. The SINR of the u th UE can be derived as

$$\gamma_u = \frac{\left| \sum_{b \in \mathcal{S}_u^{\text{BS}}} (\mathbf{h}_{b,u}^H + \mathbf{h}_{R,u}^H \Phi \mathbf{H}_{b,R}) \mathbf{f}_{b,u} \right|^2}{\sum_{i \in \mathcal{U}, i \neq u} \left| \sum_{b \in \mathcal{S}_i^{\text{BS}}} (\mathbf{h}_{b,u}^H + \mathbf{h}_{R,u}^H \Phi \mathbf{H}_{b,R}) \mathbf{f}_{b,i} \right|^2 + \sigma^2}. \quad (6)$$

Then, the achievable rate of the u th UE is expressed as

$$R_u = \log_2(1 + \gamma_u). \quad (7)$$

B. Problem Description

In the considered RIS-aided multi-BS multi-UE mmWave system, BS-UE association (user association), transmit beamforming, and RIS phase shifts all have significant impacts on the system performance and are highly interdependent. Therefore, their joint design is essential. Moreover, ensuring BS load balancing is vital for improving the overall user communication experience and facilitating carbon-aware network operation for sustainable sixth-generation (6G) systems [30]. Our objective is to jointly optimize user association, transmit beamforming, and RIS phase shifts to maximize the sum rate while satisfying the BS load balancing constraint. The problem is mathematically formulated as follows:

$$\max_{\{\delta_{b,u}\}, \{\mathbf{f}_{b,u}\}, \Phi} \sum_{u \in \mathcal{U}} R_u \quad (8a)$$

$$\text{s.t. } \delta_{b,u} \in \{0, 1\}, \quad \forall b \in \mathcal{B}, u \in \mathcal{U}, \quad (8b)$$

$$|\phi_m| = 1, \quad \forall m \in \{1, \dots, M\}, \quad (8c)$$

$$\sum_{b \in \mathcal{B}} \delta_{b,u} = 1, \quad \forall u \in \mathcal{U}, \quad (8d)$$

$$\sum_{u \in \mathcal{C}_b^{\text{UE}}} \delta_{b,u} \leq Z_{\max,b}, \quad \forall b \in \mathcal{B}. \quad (8e)$$

$$\sum_{u \in \mathcal{U}} \delta_{b,u} \|\mathbf{f}_{b,u}\|^2 \leq P_{\max,b}, \quad \forall b \in \mathcal{B}. \quad (8f)$$

Here, (8b) enforces a binary BS-UE association indicator, and (8c) imposes the unit-modulus constraint on RIS phase shifts. This work considers the case where each UE can be served by only one BS, as specified by (8d). The load-balancing constraint in (8e) limits the number of UEs served by the b th BS to $Z_{\max,b}$, and (8f) enforces the transmit power constraint, with $P_{\max,b}$ denoting the maximum transmit power allowed at the b th BS.

This complex and non-convex problem poses challenges for conventional optimization and AO-based algorithms, which are often computationally intensive and time-consuming. Similarly, traditional DNN and DRL methods face scalability limitations. To address these challenges, we develop a GNN model with inherent scalability and complexity advantages.

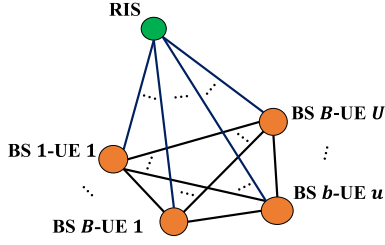


Fig. 2. Graph representation of the considered RIS-aided multi-BS multi-UE system.

The existing GNN-based methods for designing transmit beamformers and RIS phase shifts, such as those in [8], [9], [10], and [11], cannot be directly applied to joint optimization involving user association due to the additional tightly coupled variables. Besides, they did not incorporate a mechanism in their solution frameworks to address the oversmoothing issue inherent in multi-layer GNN architectures. We are therefore motivated to propose a novel GNN model to solve Problem (8), with built-in mechanisms to mitigate oversmoothing, as presented in the next section.

III. THE PROPOSED GNN-BASED APPROACH

In this section, we present the proposed GNN with adaptive attention-based aggregation method, termed GNN-AAA, to jointly optimize user association, transmit beamforming, and RIS phase adjustments for solving Problem (8), as detailed below.

A. Graph Representation

We consider a fully-connected graph consisting of one RIS node and $B \times U$ BS-UE nodes, which represent the RIS and the transmission links between B BSs and U UEs, respectively, as depicted in Fig. 2. Node features are exchanged and updated across multiple GNN layers, enabling the model to learn the system's channel and interference information. Specifically, RIS nodes learn the phase shifts, while BS-UE nodes learn the user association and beamformers.

B. Input of the GNN

The inputs of the proposed GNN model are the decorrelated received uplink pilot signals at all the BSs. Let the pilot transmitted by the u th UE be $\mathbf{x}_u \in \mathbb{C}^{L \times 1}$, where L is the length of each pilot sequence and $\|\mathbf{x}_u\|^2 = 1$. Then, the $N_T \times L$ received pilot matrix observed at the b th BS is given by

$$\mathbf{Y}_b = \sum_{u \in \mathcal{U}} (\mathbf{h}_{b,u}^H + \mathbf{h}_{R,u}^H \Phi \mathbf{H}_{b,R})^T \mathbf{x}_u^H + \mathbf{N}_b, \quad (9)$$

where $\mathbf{N}_b$ is a noise matrix. Assuming that pilots transmitted by different UEs are mutually orthogonal, the decorrelated received pilot for the u th UE at the b th BS is

$$\begin{aligned} \tilde{\mathbf{y}}_{b,u} &= \mathbf{Y}_b \mathbf{x}_u \\ &= \mathbf{h}_{b,u}^* + \mathbf{H}_{b,R}^T \Phi \mathbf{h}_{R,u}^* + \mathbf{n}_{b,u}, \end{aligned} \quad (10)$$

where $\mathbf{n}_{b,u} = \mathbf{N}_b \mathbf{x}_u$. All the channel information is contained in $\tilde{\mathbf{y}}_{b,u}$ for $b \in \mathcal{B}, u \in \mathcal{U}$, which serves as the input of the GNN without requiring explicit CSI knowledge.

In the case where pilots are not perfectly orthogonal, inter-user correlation introduces additional interference in the decorrelated received pilots:

$$\begin{aligned} \tilde{\mathbf{y}}_{b,u} &= \mathbf{h}_{b,u}^* + \mathbf{H}_{b,R}^T \Phi \mathbf{h}_{R,u}^* \\ &+ \sum_{u' \in \mathcal{U}, u' \neq u} (\mathbf{h}_{b,u'}^H + \mathbf{h}_{R,u'}^H \Phi \mathbf{H}_{b,R})^T \mathbf{x}_{u'}^H \mathbf{x}_u + \mathbf{n}_{b,u}, \end{aligned} \quad (11)$$

where $\mathbf{x}_{u'}^H \mathbf{x}_u$ captures the correlation between the u th UE and the other UEs. If a random sequence is used, where each element of $\mathbf{x}_u$ is either $-1/\sqrt{L}$ or $1/\sqrt{L}$ with equal probability, the correlation can be modeled as a Gaussian distribution with zero mean and variance $1/L$ for large L from the central limit theorem [31], [32]. If a deterministic pilot sequence is used, such as Zadoff-Chu (ZC) sequences, the correlation will be a deterministic value [33], [34].

C. GNN Structure

The structure of the proposed GNN model is shown in Fig. 3, which consists of three parts: the initial layer, Q updating layers (the first two parts are described in this subsection), and the output layer (described in Section III-E). The graph can be used to facilitate understanding of the feature exchanges in each GNN layer, as will be elaborated below. Anatomical details of Fig. 3 are given in Fig. 4.

Initial Layer: In the initial layer, as shown in Fig. 4(a), the decorrelated pilot signals $\tilde{\mathbf{y}}_{b,u}, \forall u \in \mathcal{U}, b \in \mathcal{B}$ are fed into the feature extractor $\Psi_U^{(0)}(\cdot) : \mathbb{R}^{2N_T \times 1} \mapsto \mathbb{R}^{K \times 1}$ in each BS-UE node, where $\Psi_U^{(0)}(\cdot)$ is implemented by a single-hidden-layer DNN with $2N_T$ input neurons, $2K$ hidden neurons, and K output neurons (hereafter denoted as a DNN of size $2N_T \times 2K \times K$), and K is an adjustable parameter. For the (b, u) th BS-UE node, the output of the feature extractor, represented as

$$\mathbf{z}_{(b,u)}^{(0)} = \Psi_U^{(0)}(\mathbf{z}_{(b,u)}^{(0)}) \in \mathbb{R}^{K \times 1}, \quad (12)$$

is passed to the RIS node in the initial layer and all the RIS and BS-UE nodes in the next layer, where $\mathbf{z}_{(b,u)}^{(0)} = [\Re\{\tilde{\mathbf{y}}_{b,u}^T\}, \Im\{\tilde{\mathbf{y}}_{b,u}^T\}]^T \in \mathbb{R}^{2N_T \times 1}$. The RIS node aggregates the BS-UE node features $\mathbf{z}_{(b,u)}^{(0)}, \forall u \in \mathcal{U}, b \in \mathcal{B}$ and produces the RIS node feature

$$\mathbf{z}_R^{(0)} = \Psi_R^{(0)}\left(f_0(\mathbf{z}_{(1,1)}^{(0)}, \dots, \mathbf{z}_{(b,u)}^{(0)}, \dots, \mathbf{z}_{(B,U)}^{(0)})\right) \in \mathbb{R}^{K \times 1} \quad (13)$$

using the feature extractor $\Psi_R^{(0)}(\cdot) : \mathbb{R}^{K \times 1} \mapsto \mathbb{R}^{K \times 1}$ implemented via a DNN of size $K \times 2K \times K$. To ensure the *permutation invariance* property of the GNN learning approach, the function $f_0(\cdot)$ is empirically chosen as an element-wise mean function, i.e., $f_0(\mathbf{z}_{(1,1)}^{(0)}, \dots, \mathbf{z}_{(b,u)}^{(0)}, \dots, \mathbf{z}_{(B,U)}^{(0)}) = \frac{1}{B \times U} \sum_{b \in \mathcal{B}} \sum_{u \in \mathcal{U}} \mathbf{z}_{(b,u)}^{(0)}$. Specifically, the phase shifts of the RIS generated by the GNN should be invariant to the ordering of the BS-UE pair of the GNN input.

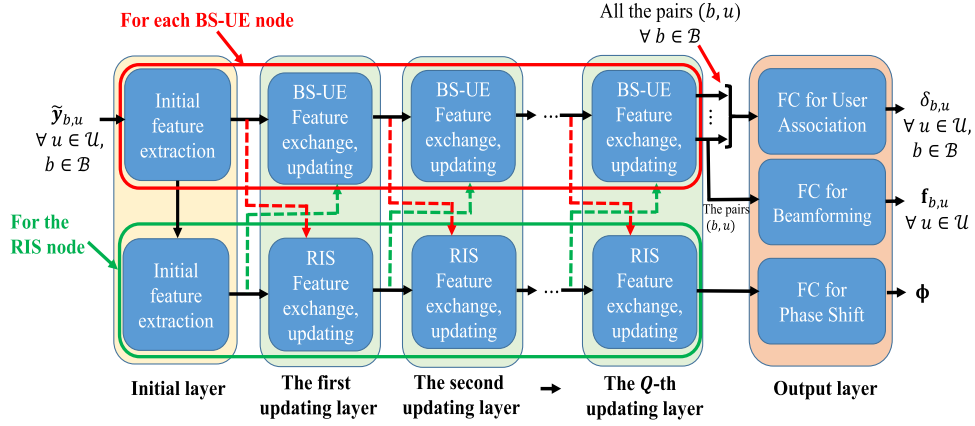


Fig. 3. The multilayered structure of the proposed GNN-AAA.

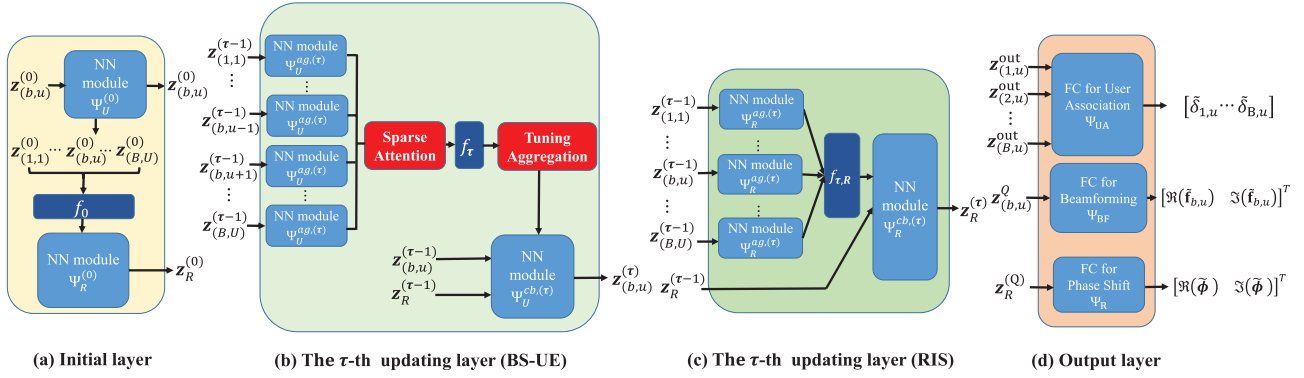


Fig. 4. Details of the proposed GNN-AAA. (a) The initial layer. (b) The τ th updating layer for the BS-UE nodes. (c) The τ th updating layer for the RIS node. (d) The output layer.

Updating Layers: In each updating layer τ ($\tau = 1, \dots, Q$), all the BS-UE nodes and the RIS node exchange information (aggregation) and update their features (combination), which are then passed to the next updating layer.

In the aggregation for the BS-UE nodes, each BS-UE node exchanges features with its neighboring BS-UE nodes. Specifically, in the τ th layer, as shown in Fig. 4(b), the feature vectors of all neighboring BS-UE nodes from the previous $(\tau - 1)$ th layer are first aggregated by $\Psi_U^{ag,(\tau)}(\cdot) : \mathbb{R}^{K \times 1} \mapsto \mathbb{R}^{K \times 1}$, implemented by a DNN of size $K \times K \times K$, and subsequently transformed into

$$\mathbf{z}_{(b,u)}^{ag,(\tau)} = f_\tau \left(\left\{ \Psi_U^{ag,(\tau)} \left(\mathbf{z}_{(d,i)}^{(\tau-1)} \right) \right\}_{(d,i) \neq (b,u)} \right) \in \mathbb{R}^{K \times 1} \quad (14)$$

to preserve the *permutation equivariance* property of the GNN. In this work, $f_\tau(\cdot)$ is empirically chosen as a summation function.

In the combination for the BS-UE nodes, each BS-UE node combines its own features from the last updating layer, the aggregated features, and the features from the RIS node, to update the node feature in the current layer. That is, the node feature vector $\mathbf{z}_{(b,u)}^{(\tau)}$ of the (b,u) th BS-UE node at the τ th layer is updated by

$$\mathbf{z}_{(b,u)}^{(\tau)} = \Psi_U^{cb,(\tau)}(\mathbf{z}_{(b,u)}^{(\tau-1)}, \mathbf{z}_{(b,u)}^{ag,(\tau)}, \mathbf{z}_R^{(\tau-1)}) \in \mathbb{R}^{K \times 1}, \quad (15)$$

where the combiner $\Psi_U^{cb,(\tau)}(\cdot) : \mathbb{R}^{3K \times 1} \mapsto \mathbb{R}^{K \times 1}$ is implemented using a DNN of size $3K \times K \times K$.

In the aggregation for the RIS nodes, all the BS-UE nodes exchange their features with the RIS node, as shown in Fig. 4(c). The aggregated BS-UE node features are given by

$$\mathbf{z}_R^{ag,(\tau)} = f_{\tau,R} \left(\left\{ \Psi_R^{ag,(\tau)}(\mathbf{z}_{(b,u)}^{(\tau-1)}) \right\}_{b \in \mathcal{B}, u \in \mathcal{U}} \right) \in \mathbb{R}^{K \times 1}, \quad (16)$$

where the aggregator $\Psi_R^{ag,(\tau)}(\cdot) : \mathbb{R}^{K \times 1} \mapsto \mathbb{R}^{K \times 1}$ is implemented using a DNN of size $K \times K \times K$, and, similar to $f_0(\cdot)$ in (13), $f_{\tau,R}(\cdot)$ is empirically chosen as an element-wise mean function to ensure permutation invariance.

In the combination for the RIS node, by combining the aggregated BS-UE node features and the RIS node features from the last layer, the feature vector of the RIS node at the τ th layer is updated by

$$\mathbf{z}_R^{(\tau)} = \Psi_R^{cb,(\tau)}(\mathbf{z}_R^{(\tau-1)}, \mathbf{z}_R^{ag,(\tau)}) \in \mathbb{R}^{K \times 1}, \quad (17)$$

where the combiner $\Psi_R^{cb,(\tau)}(\cdot) : \mathbb{R}^{2K \times 1} \mapsto \mathbb{R}^{K \times 1}$ is implemented using a DNN of size $2K \times K \times K$.

To address the oversmoothing issue caused by repeated aggregation in multi-layer GNN architectures, which can degrade the learning performance [26], we propose a mechanism for aggregation and combination, as described next.

D. Adaptive Attention-Based Aggregation

Oversmoothing refers to the issue where graph node features become increasingly similar after traversing a deep updating structure, which could potentially degrade a GNN's learning capability [26]. We propose a mechanism called adaptive attention-based aggregation (AAA) to mitigate the oversmoothing problem in GNNs, which consists of two complementary components: 1) sparse attention for aggregation, and 2) tuning aggregation for combination. Sparse attention focuses on the most important neighboring BS-UE nodes by restricting the set of neighbor BS-UE nodes with which each node exchanges information. Tuning aggregation controls whether aggregated neighbor features are incorporated into a node's updated features, especially when sparse attention alone is insufficient to suppress harmful neighbors. These harmful neighbors are those whose aggregated features make a node more similar to others, thereby exacerbating the oversmoothing effect. Adaptive and flexible adjustments are achieved through these two components to prevent excessive information loss, thereby preserving the GNN's learning capability.

1) *Sparse Attention for Aggregation*: Sparse attention for aggregation comprises the following steps: a) computing attention weights, b) sparsifying the attention matrix, and c) adaptively adjusting the sparsity level.

a) *Computing Attention Weights*: In sparse attention, the features from the neighbor nodes are first assigned attention weights. The aggregation for the BS-UE node in the τ th layer with attention is given by

$$\mathbf{z}_{(b,u)}^{\text{ag},(\tau)} = f_{\tau} \left(\left\{ \hat{a}_{(b,u),(d,i)}^{(\tau)} \Psi_U^{\text{ag},(\tau)} \left(\mathbf{z}_{(d,i)}^{(\tau-1)} \right) \right\}_{(d,i) \neq (b,u)} \right) \in \mathbb{R}^{K \times 1}, \quad (18)$$

where $\hat{a}_{(b,u),(d,i)}^{(\tau)}$ is the attention weight of the node pair between the (b,u) th BS-UE node and the (d,i) th BS-UE node in the τ th layer. The attention weight is calculated using the Euclidean distance between the features of the (b,u) th BS-UE node and the (d,i) th BS-UE node, where a larger distance corresponds to a higher attention weight, i.e.,

$$\hat{a}_{(b,u),(d,i)}^{(\tau)} = \eta \left(\left\| \mathbf{z}_{(b,u)}^{(\tau)} - \mathbf{z}_{(d,i)}^{(\tau)} \right\| \right), \quad (19)$$

where $\eta(\cdot)$ is the softmax function for normalization. The use of attention weights ensures that neighbor nodes with less similar features (and thus more informative) are assigned higher aggregation weights.

b) *Sparsifying the Attention Matrix*: A sparsity mechanism is applied to the attention weights to mitigate the oversmoothing problem. Inspired by [35], we extract only the most significant neighboring node features for information exchange, discarding the less relevant ones. First, define $\mathbf{U}^{(\tau)} \in \mathbb{R}^{BU \times BU}$ with elements $\mathbf{U}_{(b,u),(d,i)}^{(\tau)} = \hat{a}_{(b,u),(d,i)}^{(\tau)}, \forall b, d \in \mathcal{B}, u, i \in \mathcal{U}$. Then, let $\mathbf{A}^{(\tau)} \in \mathbb{R}^{BU \times BU}$ be the sparse attention weight matrix at the τ th updating layer, and let $\mathbf{A}_{(b,u)}^{(\tau)} = [a_{(b,u),(1,1)}^{(\tau)}, a_{(b,u),(1,2)}^{(\tau)}, \dots, a_{(b,u),(B,U)}^{(\tau)}] \in \mathbb{R}^{1 \times BU}, \forall b \in \mathcal{B}, u \in \mathcal{U}$ be the (b,u) th row of $\mathbf{A}^{(\tau)}$, which represents the attention weights for the (b,u) th BS-UE node

at the τ th updating layer. We find a sparse attention weight matrix by solving

$$\hat{\mathbf{A}}^{(\tau)} = \arg \min_{\mathbf{A}^{(\tau)}} \left\| \mathbf{A}^{(\tau)} - \mathbf{U}^{(\tau)} \right\|_F, \quad (20a)$$

$$\text{s.t. } \left\| \mathbf{A}_{(b,u)}^{(\tau)} \right\|_0 \leq \beta_{(b,u)}^{(\tau)} BU, \forall b \in \mathcal{B}, u \in \mathcal{U}. \quad (20b)$$

Here, (20b) represents the sparsity constraint, restricting the number of nonzero elements in $\mathbf{A}_{(b,u)}^{(\tau)}$ to $\beta_{(b,u)}^{(\tau)} BU$, where $\beta_{(b,u)}^{(\tau)} \in (0, 1]$ is a parameter that controls the degree of sparsity. By finding the optimal $\hat{\mathbf{A}}^{(\tau)}$ in (20), we retain neighbor node features that differ from a node's dominant features, enabling effective information exchange with neighbors that contribute more unique information. This mitigates oversmoothing, where nodes become overly similar after multiple rounds of information exchange. To reduce complexity, we retain only the largest $\beta_{(b,u)}^{(\tau)} BU$ elements in $\mathbf{A}_{(b,u)}^{(\tau)}$ and set the remaining elements to zero.

c) *Adaptively Adjusting the Sparsity Level*: An adaptive adjustment of $\beta_{(b,u)}^{(\tau)}$ is further applied to sparse attention. To guide this adjustment, the *power of node differences*, which is a measure of oversmoothing, is computed based on the squared differences between the feature vectors of each node pair. Specifically, for the (b,u) th node in the τ th updating layer, the power of node differences is given by

$$p_{(b,u)}^{(\tau)} = \sum_{(d,i) \in \mathcal{N}_{(b,u)}} \left\| \mathbf{z}_{(b,u)}^{(\tau)} - \mathbf{z}_{(d,i)}^{(\tau)} \right\|^2, \quad (21)$$

where $\mathcal{N}_{(b,u)}$ is the set of neighbor nodes for the (b,u) th node. A smaller power of node differences suggests a more severe oversmoothing issue for the node. Then, we compare each node's power of node differences between the τ th and $(\tau - 1)$ th updating layers and identify the nodes where the power of node differences in the τ th layer is lower than that in the $(\tau - 1)$ th layer. These nodes, which experience worsening oversmoothing across layers, are collected into the set $\mathcal{V}^{(\tau)}$. For each node in $\mathcal{V}^{(\tau)}$, we define $\beta_{(b,u)}^{(\tau)}$ based on the ratio of the decrease in each node's power of node differences relative to all nodes in $\mathcal{V}^{(\tau)}$, using the softmax function, i.e.,

$$\beta_{(b,u)}^{(\tau)} = \frac{e^{|p_{(b,u)}^{(\tau-1)} - p_{(b,u)}^{(\tau)}|}}{\sum_{(d,i) \in \mathcal{V}^{(\tau)}} e^{|p_{(d,i)}^{(\tau-1)} - p_{(d,i)}^{(\tau)}|}}, \quad \forall (b,u) \in \mathcal{V}^{(\tau)}. \quad (22)$$

For nodes not in $\mathcal{V}^{(\tau)}$, we set $\beta_{(b,u)}^{(\tau)} = 1, \forall (b,u) \notin \mathcal{V}^{(\tau)}$, as they are deemed not experiencing significant oversmoothing in the τ th updating layer.

2) *Tuning Aggregation for Combination*: In tuning aggregation, we adjust the impact of the aggregated feature on the updated feature. For each node, its oversmoothing status is evaluated. If a node exhibits worse oversmoothing in the current layer compared to the previous layer, the extent of utilizing aggregated information for that node is reduced. With the adaptive tuning aggregation, the node feature vector $\mathbf{z}_{(b,u)}^{(\tau)}$ of the (b,u) th BS-UE node at the τ th layer is updated by

$$\mathbf{z}_{(b,u)}^{(\tau)} = \Psi_U^{\text{cb},(\tau)} \left(\mathbf{z}_{(b,u)}^{(\tau-1)}, \beta_{(b,u)}^{(\tau)} \mathbf{z}_{(b,u)}^{\text{ag},(\tau)}, \mathbf{z}_R^{(\tau-1)} \right) \in \mathbb{R}^{K \times 1}, \quad (23)$$

where the aggregated feature $\mathbf{z}_{(b,u)}^{\text{ag},(\tau)}$ is adaptively weighted through the parameter $\beta_{(b,u)}^{(\tau)}$ specified in (22).

E. Output of the GNN

The output layer consists of three blocks which produce the learned results of user association, transmit beamforming, and RIS phase shifts, as shown in Fig. 4(d).

UA Block: The UA block contains U modules that generate the UA results of the U UEs, where each module is implemented by a single fully-connected (FC) layer with softmax as the output function. Specifically, we collect the features associated with the u th UE after the Q th updating layer and transform them into the feature

$$\mathbf{z}_{(b,u)}^{\text{out}} = [\mathbf{z}_{(b,u)}^{(Q)}, \mathbf{z}_u^{(Q)}]^T \in \mathbb{R}^{2K \times 1}, \quad (24)$$

where $\mathbf{z}_u^{(Q)} = \sum_{b=1}^B \mathbf{z}_{(b,u)}^{(Q)}$ can be seen as the synthesized feature associated with the u th UE. The u th UA module first maps the features $\mathbf{z}_{(b,u)}^{\text{out}}$ to a scalar $\omega_{(b,u)}$ using an FC linear function $\Psi_{\text{UA}}(\cdot) : \mathbb{R}^{2K \times 1} \mapsto \mathbb{R}$, i.e., $\omega_{(b,u)} = \Psi_{\text{UA}}(\mathbf{z}_{(b,u)}^{\text{out}})$. Then, the softmax function is applied to $\omega_{(b,u)}$ to produce B softmax values

$$\tilde{\delta}_{b,u} = \frac{e^{\omega_{(b,u)}}}{\sum_{b \in \mathcal{B}} e^{\omega_{(b,u)}}}, \quad \text{for } b = 1, \dots, B. \quad (25)$$

Since the softmax output can be interpreted as the posterior probability of association between the b th BS and the u th UE under the current channel condition, the final UA result that satisfies constraint (8d) for the u th UE can be chosen as the BS that yields the maximum $\delta_{b,u}$, i.e.,

$$\begin{cases} \delta_{b_u^*,u} = 1, & \text{for } b_u^* = \arg \max_{b \in \mathcal{B}} \tilde{\delta}_{b,u} \\ \delta_{b,u} = 0, & \text{for } b \neq b_u^*, b \in \mathcal{B}. \end{cases} \quad (26)$$

Beamformer Block: The beamformer block contains U modules that generate the transmit beamformers for all the U UEs, with the u th module of the b th BS being implemented by a single FC layer $\Psi_{\text{BF}}(\cdot) : \mathbb{R}^{K \times 1} \mapsto \mathbb{R}^{2N_T \times 1}$ to produce

$$\mathbf{z}_{(b,u)}^{\text{BF}} = \Psi_{\text{BF}}(\mathbf{z}_{(b,u)}^{(Q)}) \in \mathbb{R}^{2N_T \times 1}, \quad (27)$$

which corresponds to the beamformer for the u th UE transmitted by the b th BS determined by the UA block. The vector of the first N_T elements $\tilde{\mathbf{f}}_{b,u,1} = [\mathbf{z}_{(b,u)}^{\text{BF}}]_{1:N_T}$ represents the real part of the beamformer, while the vector of the last N_T elements $\tilde{\mathbf{f}}_{b,u,2} = [\mathbf{z}_{(b,u)}^{\text{BF}}]_{(N_T+1):2N_T}$ corresponds to the imaginary part of the beamformer. The output beamformer of the b th BS for the u th UE is then given by

$$\tilde{\mathbf{f}}_{b,u} = \tilde{\mathbf{f}}_{b,u,1} + j\tilde{\mathbf{f}}_{b,u,2}. \quad (28)$$

To satisfy the constraint (8f), a soft beamforming matrix $\tilde{\mathbf{g}}_b$ for the b th BS is constructed during training by

$$\tilde{\mathbf{f}}_b \triangleq [\tilde{\delta}_{b,1}\tilde{\mathbf{f}}_{b,1}, \tilde{\delta}_{b,2}\tilde{\mathbf{f}}_{b,2}, \dots, \tilde{\delta}_{b,U}\tilde{\mathbf{f}}_{b,U}], \quad (29)$$

which can be regarded as the contribution of beamformers from the b th BS for each UE under the soft user association $\tilde{\delta}_{b,u}, \forall u \in \mathcal{U}$ in the training stage. The virtual composite

beamformer complying with constraint (8f) for the u th UE is

$$\mathbf{f}_{b,u} = \frac{\tilde{\mathbf{f}}_{b,u}}{\sum_{b \in \mathcal{B}} \tilde{\delta}_{b,u} \|\tilde{\mathbf{f}}_b\|} \sqrt{P_{\max,b}}. \quad (30)$$

This suggests that, during testing, the u th UE's beamformer transmitted by the b_u^* th BS satisfying the power constraint (8f) is given by

$$\mathbf{f}_{b_u^*,u} = \frac{\tilde{\mathbf{f}}_{b_u^*,u}}{\sum_{i \in \mathcal{C}_{b_u^*}^{\text{UE}}} \delta_{b_u^*,i} \|\tilde{\mathbf{f}}_{b_u^*,i}\|} \sqrt{P_{\max,b_u^*}}, \quad (31)$$

where $\mathcal{C}_{b_u^*}^{\text{UE}} = \{i \mid \delta_{b_u^*,i} = 1, i \in \mathcal{U}\}$ is the set of UEs being served by the BS b_u^* that is associated with the u th UE during testing, with b_u^* defined in (26).

RIS Phase Shift Block: Finally, the RIS phase shift block collects the RIS feature vector from the Q th updating layer and generates a $2M \times 1$ vector $\mathbf{z}^{\text{RIS}} = \Psi_{\text{RIS}}(\mathbf{z}_R^{(Q)}) \in \mathbb{R}^{2M \times 1}$, where $\Psi_{\text{RIS}}(\cdot) : \mathbb{R}^{K \times 1} \mapsto \mathbb{R}^{2M \times 1}$ is implemented by an FC layer. The first M elements and the last M elements of $\mathbf{z}^{\text{RIS}}$ are the real and imaginary parts of the RIS phase shifts, respectively, and normalization is applied to ensure unit length. That is,

$$\begin{aligned} \tilde{\phi}_m &\triangleq [\mathbf{z}^{\text{RIS}}]_m + j \cdot [\mathbf{z}^{\text{RIS}}]_{M+m} \quad \text{and} \quad \phi_m = \frac{\tilde{\phi}_m}{|\tilde{\phi}_m|}, \\ m &= 1, \dots, M. \end{aligned} \quad (32)$$

F. Training of the GNN

The proposed GNN model employs *unsupervised* offline training, utilizing training samples generated from different pilot signal realizations. The training loss function combines the sum-rate maximization objective with a penalty term that enforces the load balancing constraint in (8e). It is defined as

$$\mathcal{L} = - \sum_{u \in \mathcal{U}} R_u + \lambda \cdot L_p, \quad (33)$$

where

$$L_p \triangleq \sum_{b \in \mathcal{B}} \max \left(0, \sum_{u \in \mathcal{C}_b^{\text{UE}}} \delta_{b,u} - Z_{\max,b} \right) \quad (34)$$

is the penalty for violating the load balancing constraint, and λ is a weight that balances the influence of the sum rate and the penalty for load-balancing violations.

The proposed method is summarized in Algorithm 1.

Remark 1: It is worthwhile to note that the proposed GNN model exhibits generalizability to varying numbers of UEs and BSs, i.e., the proposed model can be trained with fewer UEs and BSs while allowing testing with larger numbers of UEs and BSs. This is achieved because all learning modules within the feature extraction and update processes are shared across BS-UE links, and the architecture and the information exchange mechanism are designed to satisfy permutation invariance and permutation equivariance properties. For example, as shown in the red solid-outlined region in Fig. 3, all BS-UE nodes share the same learning module chain. Besides, the NN modules in Fig. 4(a) and Fig. 4(b) can be simply replicated multiple times when there are more UEs

Algorithm 1 The Proposed GNN-AAA Method**Input:** The decorrelated received pilot, $\tilde{\mathbf{y}}_{b,u}, \forall b \in \mathcal{B}, u \in \mathcal{U}$.**Output:** The indicator of user association, $\delta_{b,u}$, the transmit beamformer, $\mathbf{f}_{b,u}$, and the matrix of RIS phase shifts, Φ , for $b \in \mathcal{B}, u \in \mathcal{U}$

- 1: Extract the initial feature, $\mathbf{z}_{(b,u)}^{(0)}, \mathbf{z}_R^{(0)}$ according to (12) and (13), respectively
- 2: **for** each updating layer $\tau \in \{1, 2, \dots, Q\}$ **do**
- 3: Each BS-UE node generates the aggregated features $\mathbf{z}_{(b,u)}^{\text{ag},(\tau)}$ according to (18), with the weights of sparse attention $\hat{a}_{(b,u),(d,i)}^{(\tau)}$ given by (20)
- 4: Each BS-UE node updates its feature $\mathbf{z}_{(b,u)}^{(\tau)}$ according to (23), with the weights of sparse attention given by (22)
- 5: The RIS node generates the aggregated feature $\mathbf{z}_R^{\text{ag},(\tau)}$ according to (16)
- 6: The RIS node updates its feature $\mathbf{z}_{(b,u)}^{(\tau)}$ according to (17)
- 7: **end for**
- 8: Generate the final feature for user association, $\mathbf{z}_{(b,u)}^{\text{out}}$, according to (24)
- 9: Generate the indicator of user association, $\delta_{b,u}$, according to (26)
- 10: Generate the transmit beamformer, $\mathbf{f}_{b,u}$, according to (30)
- 11: Generate the matrix of RIS phase shifts, Φ , according to (32)

and/or BSs. Furthermore, the parameter settings of the learning modules in the output layer are independent of the numbers of UEs and BSs. This generalizability across varying numbers of UEs and BSs facilitates low-complexity training with fewer UEs and BSs, enables the direct application of shared, pre-trained GNN learning modules to newly added BS-UE links without retraining, and is a highly desirable property in practical dynamic network environments.

IV. COMPLEXITY ANALYSIS

In this section, we analyze the computational complexity of the proposed GNN-AAA method. The total complexity results from the sum of computations in the initial layer, updating layers, and output layer. The analysis can be expressed in terms of the number of UEs, U , the number of BSs, B , the length of a feature vector in each updating layer, K , the number of transmit antennas at each BS, N_T , the number of updating layers, Q , and the number of RIS elements, M .

Initial Layer: The computations involve input feature extraction for each BS-UE node in (12) and feature aggregation and extraction for the RIS node in (13). The computations of input feature extraction by $\Psi_U^{(0)}$ of size $2N_T \times 2K \times K$ in (12) include multiplications with neuron weights and ReLU activation, which require $\mathcal{O}(2K^2 + 4N_T K)$ and $\mathcal{O}(3K)$ operations for each BS-UE node, respectively. The complexity for all BS-UE nodes is therefore $\mathcal{O}(BU(2K^2 + (4N_T + 3)K))$. The computations in (13) include computing the element-wise mean of the BS-UE node input features, which requires $\mathcal{O}(BUK)$ operations, followed by feature extraction using

$\Psi_R^{(0)}$ of size $K \times 2K \times K$ with ReLU activation, which requires $\mathcal{O}(4K^2 + 3K)$ operations. Summing these up, the overall complexity of the initial layer is approximated as $\mathcal{O}(BU(2K^2 + (4N_T + 4)K))$.

Updating Layers: The computations for the BS-UE nodes and the RIS node are analyzed separately. The BS-UE node complexity arises from executing the AAA mechanism, as detailed below.

1) Sparse Attention for Aggregation:

The complexity encompasses three main steps: a) computing attention weights, b) sparsifying the attention matrix, and c) adaptively adjusting the sparsity level. For analytical convenience, we consider the worst-case scenario in terms of complexity, where $\beta_{(b,u)}^{(\tau)} = 1, \forall b \in \mathcal{B}, u \in \mathcal{U}$ and $\tau \in \{1, \dots, Q\}$.

a) *Computing Attention Weights:* The complexity involves aggregating features from neighboring nodes and multiplying the aggregated features by the attention weights as in (18), and computing the attention weights themselves as in (19). The computations in (18) include, for each BS-UE node, extracting features from $BU - 1$ neighboring nodes using $\Psi_U^{\text{ag},(\tau)}$ of size $K \times K \times K$ with ReLU activation, which requires $\mathcal{O}(2K^2 + 2K)$ operations, and multiplying the aggregated features by the attention weights followed by aggregation, which requires $\mathcal{O}((BU - 1)K)$ and $\mathcal{O}((BU - 2)K)$ operations, respectively. In (19), computing pairwise Euclidean distances between all BU nodes' feature vectors involves $\mathcal{O}((BU - 1)K)$ operations, and the softmax operation adds $\mathcal{O}(BU)$ operations. Overall, the total complexity for all BS-UE nodes is approximated as $\mathcal{O}(B^2U^2(3K + 1) + BU(2K^2 + 2K))$.

b) *Sparsifying the Attention Matrix:* The complexity of solving (20) is determined by projecting each row vector of $\mathbf{A}^{(\tau)}$ onto an ℓ_0 unit ball [35], which can be implemented by partial sorting and simplex projection, requiring $\mathcal{O}(BU \log(BU - 1))$ operations. Therefore, the overall complexity for all BS-UE nodes is approximately $\mathcal{O}(B^2U^2 \log(BU))$.

c) *Adaptively Adjusting the Sparsity Level:* The complexity of this step involves computing pairwise feature differences between each BS-UE node and its $BU - 1$ neighboring nodes as in (21), which requires $\mathcal{O}(B^2U^2K)$ operations, and computing the power differences with a row-wise softmax over vectors of length BU as in (22), which require $\mathcal{O}(B^2U^2)$ operations, for all BS-UE nodes. Therefore, the total complexity for all BS-UE nodes is approximated as $\mathcal{O}(B^2U^2(K + 1))$.

2) Tuning Aggregation for Combination:

The complexity of this step includes that for first performing element-wise multiplication between each BS-UE node's aggregated feature vector and the adaptive tuning adjustment as in (23), which requires $\mathcal{O}(K)$ operations, followed by feature combination using $\Psi_U^{\text{cb},(\tau)}$ of size $3K \times K \times K$ with ReLU activation, which requires $\mathcal{O}(4K^2 + 2K)$ operations. Thus, the total complexity for all BS-UE nodes is approximated as $\mathcal{O}(BU(4K^2 + 3K))$.

Summing up these complexity orders, the total complexity for the BS-UE nodes across the Q updating layers is $\mathcal{O}(QB^2U^2(\log(BU) + 4K + 2) + QBU(6K^2 + 5K))$.

The RIS node complexity involves aggregating features from all BS-UE nodes as in (16) and combining the aggregated feature with the RIS node's feature from the previous layer as in (17). In (16), feature aggregation using $\Psi_R^{\text{ag},(\tau)}$ of size $K \times K \times K$ with ReLU activation requires $\mathcal{O}(BU(2K^2 + 2K))$ operations, while the element-wise mean operation requires $\mathcal{O}(BUK)$ operations. In (17), the combination using $\Psi_R^{\text{cb},(\tau)}$ of size $2K \times K \times K$ with ReLU activation requires $\mathcal{O}(3K^2 + 2K)$ operations. Summing these up, the total complexity for the RIS node across the Q updating layers is $\mathcal{O}(QBU(2K^2 + 3K) + Q(3K^2 + 2K))$.

Output Layer: The computations include UA, beamforming, and RIS phase shift calculations for the output. The UA complexity comprises computing the FC linear function $\Psi_{\text{UA}}(\cdot)$, which requires $\mathcal{O}(2BK)$ operations, and the softmax in (25), which requires $\mathcal{O}(B)$ operations, for one UE. This sums up to $\mathcal{O}(BU(2K + 1))$ for all UEs. The beamforming complexity comprises computing $\Psi_{\text{BF}}(\cdot)$ in (27), which requires $\mathcal{O}(2N_TK)$ operations, followed by normalization to satisfy the power constraint as in (30), which requires $\mathcal{O}(N_T)$ operations, for each BS-UE node. This sums up to $\mathcal{O}(BUN_T(2K + 1))$ for all BS-UE nodes. The RIS phase shift complexity involves computing $\Psi_{\text{RIS}}(\cdot)$, which requires $\mathcal{O}(2MK)$ operations, followed by normalization to enforce the unit-modulus constraint as in (32), which requires $\mathcal{O}(M)$ operations. Adding up these numbers, the total complexity for the output layer is $\mathcal{O}(BU((2N_T + 2)K + (N_T + 1)) + M(2K + 1))$.

In summary, the overall computational complexity of the proposed GNN-AAA method is given by

$$\underbrace{\mathcal{O}(BU(2K^2 + 4N_TK + 4K))}_{\text{initial layer}} + \underbrace{\mathcal{O}(QB^2U^2(\log(BU) + 4K + 2) + QBU(8K^2 + 8K))}_{\text{updating layers}} + \underbrace{\mathcal{O}(BU((2N_T + 2)K + (N_T + 1)) + M(2K + 1))}_{\text{output layer}}. \quad (35)$$

It is worthwhile to examine the complexity of the proposed AAA mechanism relative to the overall model, excluding the computations of aggregators and combiners since they are not part of the AAA mechanism, as summarized below:

$$\underbrace{\mathcal{O}(QB^2U^2(\log(BU) + 3K))}_{\text{sparse attention}} + \underbrace{\mathcal{O}(QBUK)}_{\text{tuning aggregation}}. \quad (36)$$

Using the simulation parameters $N_T = 8$, $K = 512$, and $Q = 8$, the proportion of AAA's computational complexity order relative to the proposed GNN-AAA model is calculated as 5.82% for $B = 3, U = 6, M = 100$, 6.85% for $B = 3, U = 6, M = 200$, 10.76% for $B = 5, U = 8, M = 100$, and 11.69% for $B = 5, U = 8, M = 200$. These results indicate that the AAA mechanism adds only a modest increase

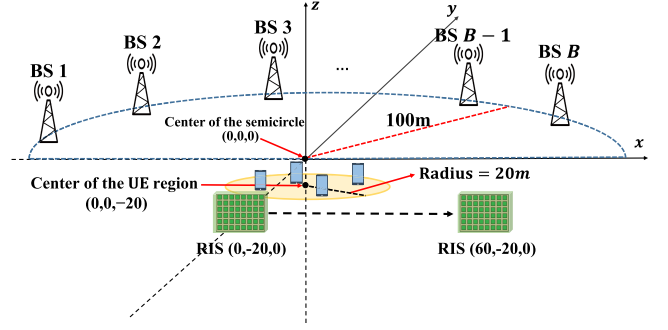


Fig. 5. System topology for simulation.

to the overall computational complexity order, which is further corroborated by runtime measurements presented in the simulation section.

V. SIMULATION RESULTS

This section presents numerical results to evaluate the performance and key features of the proposed method from various perspectives.

A. Simulation Setup

We simulate a RIS-aided multi-UE multi-BS system where each BS has $N_T = 8$ antennas and operates in the $f_c = 28$ GHz mmWave band. For training, the default parameters are set as $B = 3$ BSs, $U = 6$ UEs, and $M = 100$ passive RIS elements. These parameters are adjusted for testing as specified in the subsequent simulation results. The BSs are randomly distributed along a semicircular arc centered at $(0,0,0)$ (in meters) in Cartesian coordinate with a radius of 100 m, and the RIS is at $(-15, -20, 0)$, as shown in Fig. 5. Unless otherwise noted, UEs are uniformly distributed in a circular region with a radius of 20 m, centered at $(0,0,-20)$. For the mmWave channel, there are $N_{cl} = 5$ clusters in each channel, with $L_\ell = 10$ rays per cluster. The azimuth/elevation AoAs and AoDs, $\psi_{\ell,i}^{b,t}, \theta_{\ell,i}^{b,t}, \psi_{\ell,i}^{R,t}, \theta_{\ell,i}^{R,t}, \psi_{\ell,i}^{R,r}, \theta_{\ell,i}^{R,r}, \forall b \in \mathcal{B}$, follow the uniform distribution over $[0, 2\pi]$. The path loss of the channel associated with the RIS is modeled as $30 + 22 \log(d_{*,R})$, where $d_{*,R}$ is the distance between the communication node $*$ (which may be a BS or a UE) and the RIS. The path loss between the b th BS and the u th UE with distance $d_{b,u}$ is modeled as $32.6 + 36.7 \log(d_{b,u})$. The pilot length is $L = 31$. The uplink pilot transmission power and the noise power are set as 15 dBm and -85 dBm, respectively. The default BS maximum transmit power is set as $P_{\max,b} = 25$ dBm for all b . For the GNN, the sizes of the neural network (NN) modules in each layer are set as follows: $2N_T \times 1024 \times 512$ for $\Psi_U^{(0)}$, $512 \times 1024 \times 512$ for $\Psi_R^{(0)}$, $512 \times 512 \times 512$ for $\Psi_U^{\text{ag},(\tau)}$, $\Psi_U^{\text{cb},(\tau)}$, $\Psi_R^{\text{ag},(\tau)}$, and $\Psi_R^{\text{cb},(\tau)}$ for $\tau = 1, 2, \dots, Q$, where $Q = 8$ is the number of updating layers in the GNN. The number of training samples in each epoch is 10^5 , with a total of 200 epochs, and the batch size is 1024. An initial learning rate is set as 10^{-3} which decreases after 3 epochs by a factor of 0.98. The number of testing samples is 10^5 . The weight λ is

TABLE III
SIMULATION PARAMETERS

Number of BSs in training	$B = 3$
Number of UEs in training	$U = 6$
Number of transmit antenna per BS	$N_T = 8$
Carrier frequency	28 GHz
Number of RIS elements (default)	$M = 100$
Number of clusters per channel	$N_{cl} = 5$
Number of rays per cluster	$L_\ell = 10$
Length of each pilot sequence	$L = 31$
Uplink pilot transmission power	15 dBm
Noise power	-85 dBm
BS maximum transmit power (default)	$P_{\max,b} = 25$ dBm, $\forall b \in \mathcal{B}$
Size of the NN modules in the initial layer	$\Psi_U^{(0)} : 2N_T \times 1024 \times 512$ $\Psi_R^{(0)} : 512 \times 1024 \times 512$,
Size of the NN modules in the updating layer	$\Psi_U^{\text{ag},(\tau)}, \Psi_U^{\text{cb},(\tau)}, \Psi_R^{\text{ag},(\tau)}, \Psi_R^{\text{cb},(\tau)} : 512 \times 512 \times 512$ for $\tau = 1, 2, \dots, Q$
Number of updating layers	$Q = 8$
Number of training samples per epoch	10^5
Number of epochs	200
Batch size	1024
Learning rate	10^{-3} , decreasing after 3 epochs by a factor of 0.98
Number of testing samples	10^5
Tunable weight in the loss function	$\lambda = 2$
Maximum number of UEs served by a BS	$Z_{\max,b} = U/B$, $\forall b \in \mathcal{B}$

set to 2 and $Z_{\max,b}$ is set to U/B for all b . All simulation parameters are summarized in Table III.

We denote the comparison schemes using the format: ‘method [remark],’ where the ‘remark’ is optional. The schemes in comparison are described as follows:

- *GNN-AAA [proposed]*: The proposed method.
- *GNN-AAA [no LB]*: Same as the proposed method but without load-balancing (LB) considerations, i.e., (34) is not included in the loss function.
- *GNN-AAA [for BF] + RSP [for UA]*: The proposed method is applied for BS and RIS beamforming (BF) only, while UA is determined by selecting the BS that provides the highest received signal power (RSP) at the UE.
- *GNN-AAA [no RIS]*: The proposed method is applied to the scenario without RIS.
- *GNN*: The GNN model without incorporating the AAA mechanism.
- *HGAT*: The heterogeneous graph attention network (HGAT) based on the GAT framework in [36] is applied to the considered problem using the same loss function as the proposed GNN, with its attention mechanism and attention weight design adapted accordingly.
- *DNN*: A conventional DNN with fully connected layers, with the same model size, input, and loss function as the proposed GNN.
- *Model-based*: The model-based method from [14], which jointly optimizes BS and RIS beamforming as well as UA.

B. Convergence

Fig. 6 plots the training loss given in (33) against the training epochs to analyze the convergence behavior of *GNN-AAA [proposed]*. The results indicate that the inclusion of the penalty term (34) for load balancing (*GNN-AAA [proposed]*)

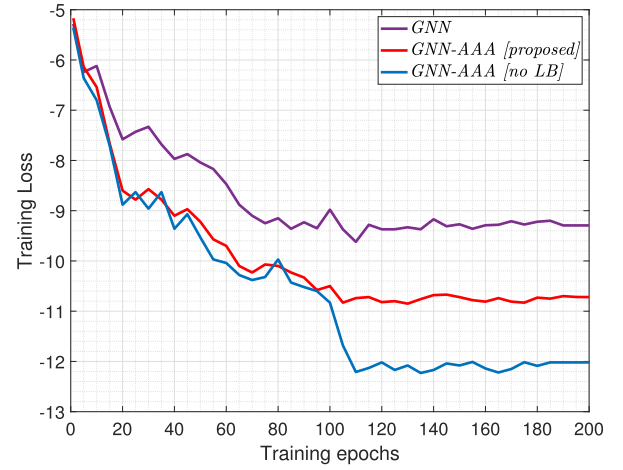


Fig. 6. Training loss vs. training epochs, where $B = 3$, $U = 6$, $P_{\max,b} = 25$ dBm $\forall b$, and $M = 100$.

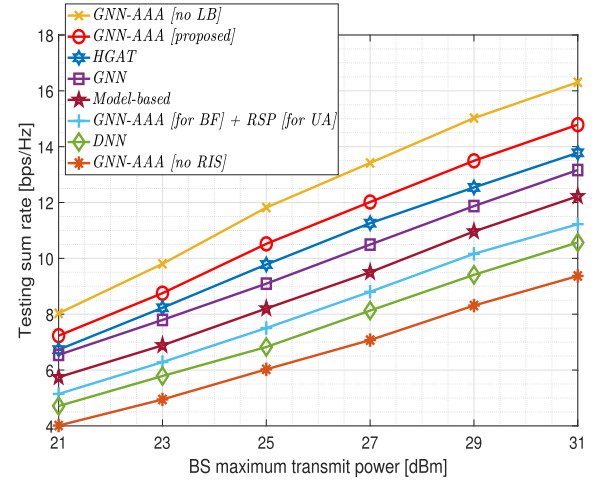


Fig. 7. Testing sum rate vs. BS maximum transmit power ($P_{\max,b}$, $\forall b$) for $B = 3$, $U = 6$, and $M = 100$.

vs. *GNN-AAA [no LB]*) and the incorporation of the AAA mechanism (*GNN-AAA [proposed]* vs. *GNN*) have minimal impact on the convergence rate. All three schemes demonstrate consistent convergence at approximately 120 training epochs. However, since the loss function is inversely related to the sum rate, they converge to different final loss values, which roughly correspond to their respective sum-rate performance, as discussed next.

C. Sum-Rate Performance

Effects of BS Transmit Power: Fig. 7 presents the testing sum-rate performance with respect to BS maximum transmit power. *GNN-AAA [no LB]* achieves the highest sum rate, although it results in a high violation of the LB constraint (8e), compared to nearly 0% for all other schemes (less than 0.1%). This highlights the effectiveness of incorporating the penalty term (34) in the loss function to meet (8e). While unbalanced UA might provide higher sum rates in certain scenarios, load-balanced UA ensures equitable resource allocation, consistent user experiences, and robust network operations. *GNN-AAA [proposed]* outperforms all benchmarks under the same LB constraints. Specifically, it achieves a higher sum rate than

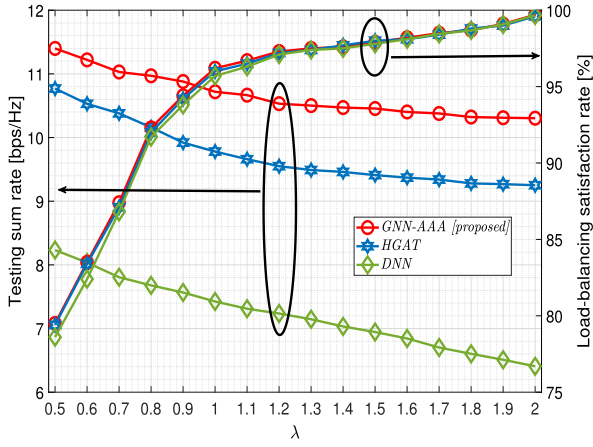


Fig. 8. Testing sum rate and load-balancing satisfaction rate vs. λ in (33) for $B = 3$, $U = 6$, $P_{\max,b} = 25$ dBm $\forall b$, and $M = 100$.

Model-based and *DNN* methods due to its superior architecture design and enhanced feature extraction capabilities. Moreover, it outperforms *HGAT*, highlighting the effectiveness of our oversmoothing-aware attention design and adaptive adjustment strategy in improving learning performance. While *HGAT* leverages an attention mechanism that enhances generalization to heterogeneous graph structures and varying neighborhood sizes, and thus yielding better performance than *GNN*, it is not specifically tailored to mitigate oversmoothing.

Among GNN-based schemes, *GNN-AAA [proposed]* outperforms *GNN*, demonstrating that the AAA mechanism improves learning performance by effectively mitigating oversmoothing. Compared to *GNN-AAA [for BF] + RSP [for UA]*, *GNN-AAA [proposed]* achieves a higher sum rate since RSP-based UA does not consider multiuser interference during UA. This demonstrates the effectiveness of *GNN-AAA [proposed]* in learning channel and interference relationships among all BS–RIS–UE links from pilot signals. Moreover, a comparison between *GNN-AAA [proposed]* and *GNN-AAA [no RIS]* shows the benefits of deploying RIS in the system.

Effects of the Weight λ in (33): Fig. 8 shows how the testing sum rate and load-balancing satisfaction rate vary with λ . As λ increases, the satisfaction rate increases while the sum rate decreases for all learning-based schemes, reflecting the loss function in (33), where higher λ prioritizes load-balancing over sum rate. At $\lambda = 2$, the satisfaction rate reaches approximately 99.9%, demonstrating the effectiveness of the proposed loss function. Thus, $\lambda = 2$ is selected for all simulations to achieve good sum rate performance while satisfying the load-balancing constraint.

Pilot Correlation Effects: Fig. 9 examines the pilot correlation effects, as modeled in (11), where UE pilots are not perfectly orthogonal. Two pilot types are considered: random and ZC sequences, both of length $L = 31$. The results show that while all methods experience sum-rate degradation under the presence of pilot correlation, *GNN-AAA [proposed]* and *HGAT* are less affected, maintaining performance close to the ideal (uncorrelated) condition. In contrast, *DNN* suffers a more noticeable performance drop, especially with the ZC sequence. This highlights the superior robustness of GNNs against pilot correlations. Furthermore, it is observed that the

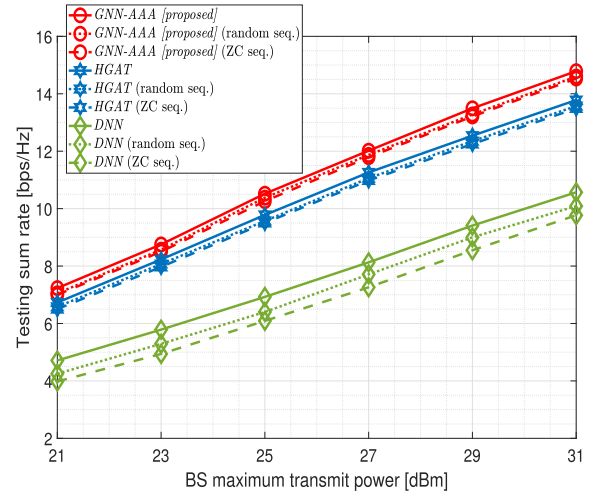


Fig. 9. Testing sum rate vs. BS maximum transmit power ($P_{\max,b}$, $\forall b$) for $B = 3$, $U = 6$, and $M = 100$, under different pilot correlation scenarios using random and ZC pilot sequences of length $L = 31$.

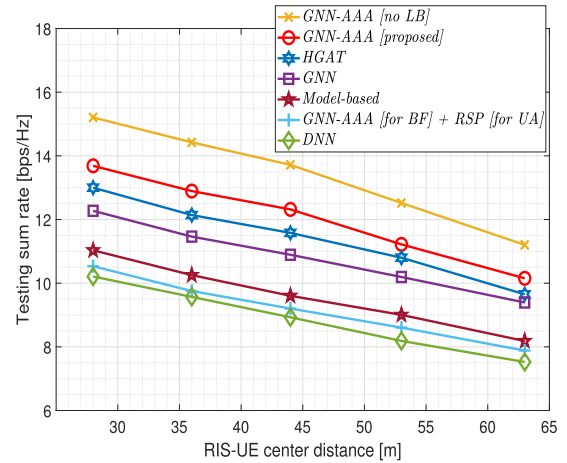


Fig. 10. Testing sum rate vs. RIS–UE center distance for $B = 3$, $U = 6$, $M = 100$, and $P_{\max,b} = 25$ dBm.

random sequence outperforms the ZC sequence across all three schemes. This difference primarily arises from the distinct types of additional correlation introduced in each case. In the random sequence case, the correlation follows a Gaussian distribution with zero mean and variance $1/L$, whereas in the ZC sequence case, it is a deterministic value of $1/\sqrt{L}$. Since the random sequence has a higher probability of producing smaller correlation values than the ZC sequence, it generally imposes less negative impact on the learning process of a data-driven model, leading to smaller degradation in sum-rate performance.

Effects of RIS–UE Distances: Fig. 10 presents the testing sum-rate performance vs. the distance between the RIS and the UE center. The UE center is fixed at $(0, 0, 0)$, while the RIS position is moved from $(0, -20, 0)$ to $(60, -20, 0)$, as shown in Fig. 5. The performance declines as the distance between the UE center and the RIS increases due to longer reflection paths and higher path loss, which reduce the received signal strength. The maximum sum rate is achieved when the RIS–UE center distance is 20 m, representing the optimal RIS position to assist all BSs along the semicircle in serv-

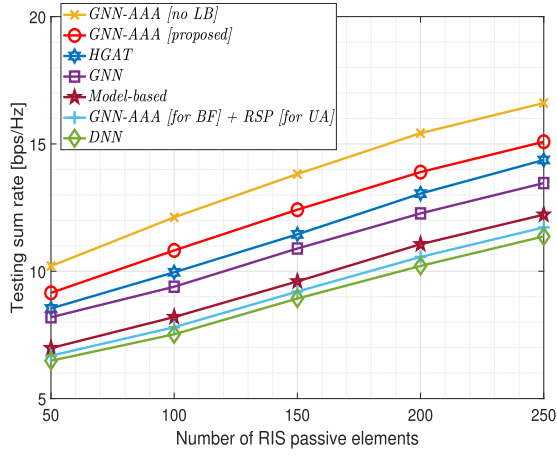


Fig. 11. Testing sum rate vs. number of RIS passive elements (M) for $B = 3$, $U = 6$, and $P_{\max,b} = 25$ dBm.

ing the UEs effectively. Furthermore, *GNN-AAA [proposed]* demonstrates adaptability to varying RIS positions, maintaining superior performance compared to other schemes. The comparative trends across all schemes remain consistent with those observed in Fig. 7.

Effects of the Number of RIS Elements: Fig. 11 presents the testing sum-rate performance vs. the number of RIS passive elements. An increasing number of RIS elements leads to higher sum rates due to enhanced received signal strength and greater flexibility to reshape the wireless channel. Notably, *GNN-AAA [proposed]* with $M = 50$ delivers performance comparable to *Model-based* and *DNN* schemes with $M = 150$, demonstrating its efficiency, as a higher number of RIS elements increases the complexity of passive beamforming design and channel estimation. The proposed GNN-AAA even avoids the need for explicit channel estimation altogether by directly using received pilots from UEs as input, which is a favorable property when adopting a higher number of RIS elements.

Discrete RIS Phases: Fig. 12 shows the testing sum rate vs. BS transmit power with 2-bit quantized RIS phase shifts, where continuous phases from each method are discretized to reflect practical RIS implementation. Compared to Fig. 7, all methods experience a performance loss, about 1.5 bps/Hz. Despite this, *GNN-AAA [proposed]* retains a consistent advantage over other benchmarks in both continuous and discrete RIS phase conditions.

D. Effects of RIS on UA

In Fig. 7, it was shown that *GNN-AAA [proposed]* achieves a significantly higher sum rate compared to *GNN-AAA [no RIS]*. This improvement stems not only from enhanced received signal strength and reduced interference, as previously mentioned, but also from the added flexibility of RIS in providing additional communication paths that help extend BS coverage and improve UA decisions. An illustrative example, comparing UA decisions by *GNN-AAA [no RIS]* and *GNN-AAA [proposed]*, is shown in Fig. 13. Different colors represent the UEs served by each BS, while gray beams indicate blocked paths. For *GNN-AAA [no RIS]*, UE 1 is associated with BS 2, and UE 3 is associated with BS 1 to satisfy the LB constraint. In contrast,

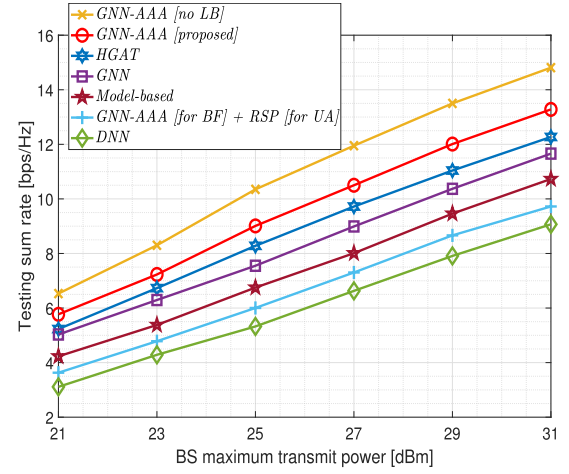


Fig. 12. Testing sum rate vs. BS maximum transmit power ($P_{\max,b}, \forall b$) for $B = 3$, $U = 6$, and $M = 100$, with 2-bit quantized RIS phase shifts.

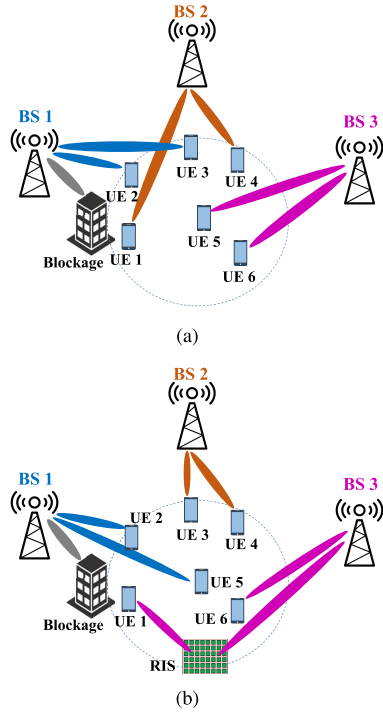


Fig. 13. Example scenarios to illustrate the effects of RIS on UA in multi-BS multi-UE systems, with $B = 3$, $U = 6$, $M = 100$, and $P_{\max,b} = 25$ dBm. (a) UA for *GNN-AAA [no RIS]*. (b) UA for *GNN-AAA [proposed]*.

for *GNN-AAA [proposed]*, UE 3 selects BS 2, its nearest BS, while UE 1 connects to BS 3 through the RIS which compensates for the path loss caused by the greater distance. This configuration leverages the RIS's flexibility to make more efficient UA decisions, resulting in higher average sum rates. Such improved UA decisions are consistently observed across testing samples, demonstrating the effectiveness of the proposed GNN-AAA in optimizing UA strategies for higher sum rates while maintaining LB considerations in RIS-aided multi-BS multi-UE systems.

TABLE IV

POWER OF NODE DIFFERENCES AND SUM RATE FOR $B = 3$, $U = 6$, $M = 100$, $Q = 8$, AND $P_{\max,b} = 25$

Scheme	Power of node differences	Sum rate (bps/Hz)
<i>GNN-AAA [proposed]</i>	9.702	10.519
<i>HGAT</i>	8.421	9.780
<i>GNN</i>	4.622	8.870
<i>GNN w/ residual connections</i>	9.203	9.987
<i>GNN-AAA w/ non-adaptive aggregation</i>	12.052	9.816

E. Effects of Oversmoothing

The power of node differences, defined in (21), is used to quantify the level of oversmoothing in GNN-based methods (higher values indicate less oversmoothing). Table IV presents the performance of *GNN-AAA [proposed]* and other benchmarks in terms of power of node differences and sum rate. Comparing *GNN-AAA [proposed]* and *GNN*, it is seen that the improved sum rate of *GNN-AAA [proposed]* is due to reduced oversmoothing. However, focusing solely on mitigating oversmoothing does not necessarily enhance sum-rate performance. To illustrate this, a variant of *GNN-AAA [proposed]* using non-adaptive aggregation, termed *GNN-AAA w/ non-adaptive aggregation*, is tested, where sparse attention and tuning aggregation are enforced for all nodes in all layers without the mechanism in (22), i.e., $\beta_{(b,u)}^{(\tau)}$ is computed for all (b,u) rather than just those in $\mathcal{V}^{(\tau)}$. While *GNN-AAA w/ non-adaptive aggregation* achieves the highest power of node differences, meaning least oversmoothing, its sum rate is lower than that of *GNN-AAA [proposed]*. This suggests that for deep GNNs, it is crucial to make judicious decisions on whether and to what extent to aggregate information among neighboring nodes. By monitoring the oversmoothing status of each BS-UE node in every updating layer and addressing it adaptively at the node and layer levels, *GNN-AAA [proposed]* avoids unnecessary sacrifices in information exchange and achieves superior overall performance.

An existing method that tackles oversmoothing by incorporating residual connections into GNNs, termed *GNN w/ residual connections* [25], is also compared. *GNN-AAA [proposed]* outperforms *GNN w/ residual connections* in both power of node differences and sum rate. This is because the residual connections force each node at every layer to combine with initial input features without controlling information exchange, making it less adaptable to individual node states and thus less effective at mitigating oversmoothing.

HGAT mitigates oversmoothing by assigning varying importance to neighboring nodes during aggregation, which helps preserve node feature uniqueness. As a result, it achieves higher power of node differences and sum rate compared to *GNN*. However, it still falls short of *GNN-AAA [proposed]*, which more effectively addresses oversmoothing through its tailored attention weights and adaptive adjustment mechanisms.

F. Complexity

Table V reports the average runtime required to obtain a feasible solution for one testing sample under different

TABLE V

RUNTIME (UNIT: MILLISECOND)

Scheme	$M = 100$	
	$B = 3, U = 6$	$B = 5, U = 8$
<i>GNN-AAA [proposed]</i>	3.28	3.72
<i>HGAT</i>	3.11	3.30
<i>GNN</i>	3.03	3.21
<i>GNN-AAA [for BF] + RSP [for UA]</i>	3.28	3.43
<i>Model-based</i>	8.66×10^3	12.76×10^3
<i>DNN</i>	3.11	3.36
<i>GNN-AAA [no RIS]</i>	2.89	3.13

Scheme	$M = 200$	
	$B = 3, U = 6$	$B = 5, U = 8$
<i>GNN-AAA [proposed]</i>	3.52	3.93
<i>HGAT</i>	3.35	3.57
<i>GNN</i>	3.25	3.45
<i>GNN-AAA [for BF] + RSP [for UA]</i>	3.47	3.63
<i>Model-based</i>	9.46×10^3	13.83×10^3
<i>DNN</i>	3.42	3.68
<i>GNN-AAA [no RIS]</i>	3.17	3.36

TABLE VI

SUM RATE FOR *GNN-AAA [proposed]* OVER VARIOUS NUMBERS OF BSS (B) IN TRAINING AND TESTING, WITH $M = 100$, $U = 6$, AND $P_{\max,b} = 25$

B (testing)	2	3	4	5
Sum rate (bps/Hz) B (training) = 3	10.116	10.519	10.977	11.445
Sum rate (bps/Hz) B (training) = B (testing)	10.259	10.519	11.202	11.782

settings of BSs (B), UEs (U), and RIS elements (M). While incorporating AAA in *GNN-AAA [proposed]* results in a slight increase in runtime compared to *GNN*, it significantly enhances sum-rate performance, as demonstrated previously. The runtime increase from incorporating the AAA mechanism in *GNN-AAA [proposed]* generally agrees with the complexity analysis in Section IV, where the AAA mechanism contributes roughly 10% of the total computational complexity. This confirms that the AAA mechanism adds minimal overhead to the overall complexity. Furthermore, although *DNN* runs slightly faster than *GNN-AAA [proposed]*, it delivers inferior sum-rate performance. The *Model-based* method has the highest runtime due to its iterative optimization process. *GNN-AAA [no RIS]* achieves the shortest runtime as it bypasses RIS phase shift calculations but suffers from the lowest sum-rate performance due to the absence of RIS. As B , U , or M increases, runtime increases for all schemes.

G. Generalizability

Table VI and Table VII evaluate the generalizability of the proposed *GNN-AAA* across different numbers of BSs and UEs, respectively, by analyzing cases with mismatched training and testing scenarios. In Table VI, the highest sum rate is achieved when the numbers of BSs used for training and testing match (bottom row). However, even with a mismatch, the proposed method maintains comparable performance with minimal degradation, highlighting its ability to be trained on fewer BSs for reduced complexity or when the exact number of BSs in testing is unknown. Table VII shows similar results across different numbers of UEs, demonstrating the

TABLE VII
SUM RATE FOR GNN-AAA [Proposed] OVER VARIOUS NUMBERS OF UES (U) IN
TRAINING AND TESTING, WITH $M = 100$, $B = 3$, AND $P_{\max,b} = 25$

U (testing)	2	3	4	5	6	7	8	9
Sum rate (bps/Hz) U (training) = 6	9.265	9.877	10.242	10.416	10.519	10.674	10.741	11.167
Sum rate (bps/Hz) U (training) = U (testing)	9.838	10.234	10.376	10.490	10.519	11.128	11.448	11.976

practicality of GNN-AAA [proposed] since the number of UEs often varies over time and is typically unknown in advance.

Note that *Model-based* and *DNN* approaches do not possess generalization properties. The *Model-based* method use optimization and AO-based algorithms to simplify or decompose the objective function, with their structure inherently tied to the number of BSs and UEs. Any change in BS or UE numbers requires rerunning these methods to find a solution. DNN-based methods rely on fully connected layers to map decorrelated pilots to system variables, with input and output structures tied to the number of BSs and UEs. Any change in BS or UE numbers requires redesigning the DNN model architecture and retraining the model.

VI. CONCLUSION

In this paper, we have proposed a GNN-AAA approach for the joint optimization of UA, BS transmit beamforming, and RIS phase shifts in RIS-aided multi-BS multi-UE mmWave networks. The proposed GNN-AAA demonstrated enhanced generalization capabilities across both numbers of BSs and UEs compared to traditional GNNs, exhibiting only minor performance degradation in mismatched training and testing scenarios. This generalizability highlights the practicality of the proposed method for dynamic and scalable network environments. The AAA mechanism integrated into GNN-AAA showed quantifiable effectiveness in mitigating oversmoothing in deep GNNs, as reflected by improvements in the power of node differences, while maintaining adequate selective information exchange to ensure superior sum-rate performance, with only a modest increase in computational complexity. Furthermore, the proposed GNN-AAA effectively bypassed the need for explicit channel estimation, addressing a critical bottleneck in large-RIS-aided systems. Simulation results validated that GNN-AAA achieved higher sum-rate performance compared to model-based method, HGAT, DNN, and conventional UA strategies from various perspectives, demonstrated GNN-AAA's robustness against pilot correlation, and highlighted the load-balancing benefits of deploying RISs.

Future research directions include exploring alternative graph representations to model complex interactions and extending the proposed framework to scenarios involving multiple RISs. These advancements could further enhance the flexibility, scalability, and performance of GNN-AAA in more general and complex network topologies.

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